TEXORA · LECTURE SHEET

Garments Dyeing

Dyeing the sewn garment instead of the fabric: dye classes for garment work, paddle and drum machines, the temperature profile, and a fully dosed reactive dyeing calculation.

AM-4101 · ADVANCED APPAREL MANUFACTURING-I · TOPIC 2 · SPECIALIZATION — APPAREL MANUFACTURING

1. Why dye garments at all

Fabric-stage dyeing commits colour months before sale. **Garment dyeing** inverts the logic: garments are sewn from prepared-for-dyeing (PFD) undyed fabric and coloured at the last responsible moment. The gains are commercial as much as technical — one greige garment stock serves every colourway, colour decisions follow actual orders, and unsold colours can be re-dyed darker. The costs are real too: seams, thread and trims must dye compatibly (polyester sewing thread does not take cotton dyes, giving deliberate or accidental contrast stitching), shrinkage happens after sewing so patterns must be cut oversize (Topic 2 of AM-3207), and shade uniformity across a sewn, three-dimensional garment is harder than across flat fabric.

2. Dye classes used on garments

DYE CLASS	MAIN FIBRE	CHARACTER IN GARMENT WORK
Reactive	Cotton, viscose	Covalent bond to cellulose; the best wash fastness; needs salt for exhaustion and alkali for fixation — the worked example below
Direct	Cotton	Simple, economical, wide shade range; moderate wet fastness, improved by after-treatment
Pigment	Any (surface binder)	Not a true dye: pigment is bound to the surface; gives washed-out, vintage colour and stiffens hand slightly; basis of many fashion garment colours
Sulphur	Cotton	Cheap blacks, browns and dark shades; tendering risk if poorly oxidised/stored
Disperse	Polyester garments	Applied at high temperature (about 130°C) or with carriers in drum machines rated for it

3. Machines

Garment dyeing runs in machines that move garments gently through the bath: **paddle machines** (a slow paddle wheels garments through a large bath — soft action, high liquor ratio) and **rotary drum / sample-to-bulk tumbler machines** (perforated rotating drums, lower liquor ratio, better economy and control; the modern standard, often combined washer-dyed-extractor units). Machine loading is by weight and bulk: overloading creases garments and patches the shade; underloading wastes water and chemical per piece — both errors appear in the costing logic of Topic 5.

4. The reactive dyeing, calculated and profiled

WORKED EXAMPLE — DOSING A 200 KG COTTON BATCH

Batch: **200 kg** of PFD cotton garments; liquor ratio **1 : 8** → bath = 200 × 8 = **1,600 L**.

Reactive dye at 2% owg = 200 × 0.02 = **4.0 kg**.

Salt (exhaustion) at 40 g/L = 1,600 × 40 = **64 kg**.

Soda ash (fixation alkali) at 15 g/L = 1,600 × 15 = **24 kg**.

Note the two currencies again (Topic 1 of this course): dye is dosed on garment weight, auxiliaries on bath volume — the batch sheet must carry both, each against its own basis.

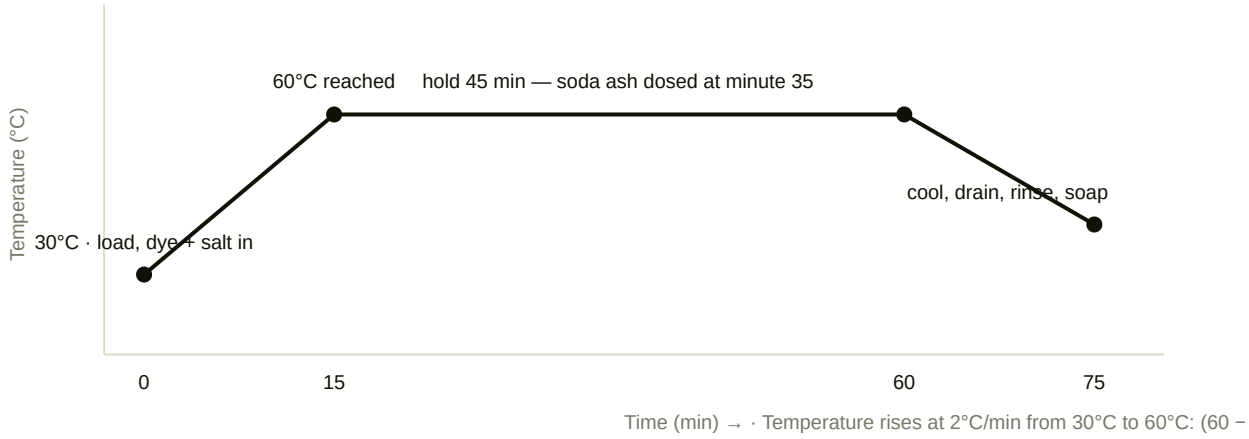


Figure 1 — Temperature profile of the worked reactive dyeing. The ramp rate controls levelness: dye strikes as temperature rises, so a controlled 2°C/min climb is a quality parameter, not a convenience.

5. Shade approval and faults

Colour is approved against the buyer's standard under standard illuminants (D65 first, then store light for metamerism), from a **lab dip** at recipe stage and a first-batch bulk sample before the run continues. The characteristic garment-dyeing faults all trace to the process: **unlevel shade** (overloading, fast ramp, late salt), **crease marks** (garments folded in the drum at high temperature, set permanently), **contrast**

stitching (polyester thread where cotton was expected — or vice versa when the design wanted it), and **trim mismatch** (buttons, zips and labels that dye differently from the body). Seam allowances and thick parts dye lighter where liquor circulation is poor — the reason garment dyeing recipes are proven on sewn trial garments, not fabric swatches.

Glossary

TERM	MEANING
PFD	Prepared for dyeing: undyed fabric/garments made to dye evenly at garment stage.
Exhaustion	Movement of dye from bath onto fibre, driven here by salt; fixation then bonds it (alkali, for reactives).
Lab dip	A small trial dyeing submitted for shade approval before bulk.
Metamerism	Two samples matching under one light source and mismatching under another.
Liquor ratio	Bath volume per garment weight (Topic 1); garment machines run lower ratios than fabric machines to save water and chemicals.

Key points and summary

- Garment dyeing trades fabric-stage commitment for late colour flexibility — at the price of compatibility and shrinkage control in the sewn garment.
- Reactive dyes are the quality standard for cotton garments: salt exhausts, alkali fixes; pigment dyeing colours the surface by binder.
- Worked batch: 200 kg at LR 1:8 needs 1,600 L, 4.0 kg dye (2% owg), 64 kg salt (40 g/L), 24 kg soda ash (15 g/L).
- The temperature profile is a calculation: 30 → 60°C at 2°C/min takes 15 min, hold 45 min, with alkali dosed mid-hold.

Practice

Answer in your own words; brief model answers follow.

1. Scale the worked recipe to a 150 kg batch at LR 1:8: bath volume, dye at 2% owg, salt at 40 g/L, soda ash at 15 g/L.
2. A dyeing is ramped from 25°C to 55°C at 1.5°C/min. How long is the ramp?

3. Why does polyester sewing thread stay white on a reactive-dyed cotton garment, and when is this a fault rather than a feature?
4. State two faults caused by overloading a garment dyeing machine and their mechanisms.

Answer key

1. Bath = 150 × 8 = 1,200 L. Dye = 150 × 0.02 = 3.0 kg. Salt = 1,200 × 40 = 48 kg. Soda ash = 1,200 × 15 = 18 kg.
2. (55 − 25) ÷ 1.5 = 20 min.
3. Reactive dyes bond to cellulose; polyester has no reactive sites for them and needs disperse dyes at high temperature. It is a feature when the design specifies contrast stitching, a fault when the buyer's standard shows matching thread.
4. Unlevel shade — liquor cannot circulate evenly through a packed load, so dye strikes unevenly. Crease marks — garments held folded at dyeing temperature set permanent creases. (Patches at thick seams are also acceptable.)

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